

FS-MPC of Microgrid Interface Converters for Reactive Power and Harmonic Compensation: A Systematic Literature Review

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Finite-Set Model Predictive Control of Microgrid Interface Converters for Reactive Power and Harmonic Compensation: A Systematic Literature Review

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Abstract — The microgrid interface converter (MIC) is rapidly evolving from a unidirectional power-routing device into a multifunctional grid-supporting asset that simultaneously delivers bulk active power, voltage regulation, reactive power compensation, and harmonic cancellation. Finite-set model predictive control (FS-MPC), with its explicit handling of switching states, fast dynamic response, and natural ability to track distorted current references, has become a leading inner-loop strategy for this multifunctional MIC. This article reviews the FS-MPC literature in the context of microgrid interface converter design, anchored by the canonical formulation of Perez and Flores-Bahamonde (2016) for reactive- and harmonic-power compensation. We follow a PRISMA-aligned protocol over IEEE Xplore, Scopus, Semantic Scholar, and Web of Science, screening 175 candidate documents (a curated Tier-1 set of 10 papers totalling 13.0 MB and 175 pages, augmented by 25 seminal/recent additions). The review (i) constructs a taxonomy of MIC control strategies; (ii) maps the chronological evolution from droop-based foundations (2002) through the FS-MPC inflection (2007–2015) to data-driven and digital-twin extensions (2024–2026); (iii) presents a correlation graph of intellectual lineage; (iv) tabulates a 12-dimensional gap analysis across the surveyed corpus; and (v) ranks 18 prospective research directions by publication relevance, patentability, and PhD-thesis strength. The review identifies energy-domain DC-link control, mode-agnostic operation, and load-side harmonic absorption as the three under-exploited levers in the current literature, and recommends multi-agent FS-MPC orchestrated by a digital twin as the highest-leverage direction for the author’s MAS-DT-SH thesis.

Index Terms — Finite-set model predictive control (FS-MPC), microgrid interface converter (MIC), reactive power compensation, harmonic compensation, active power filter, voltage source inverter (VSI), digital twin, multi-agent system.

I. INTRODUCTION

A. Motivation

Distributed energy resources (DERs) connected at low-voltage distribution levels are reshaping the operational paradigm of power systems. Microgrids — locally-bounded clusters of generation, storage, and load — provide a structural answer to the integration question, but their value to the wider grid is realised only when their grid-tied power-electronic interface contributes actively to grid services [1], [4]. The microgrid interface converter (MIC) — typically a three-phase two-level voltage source inverter (VSI) at the point of common coupling (PCC) — has therefore become a central object of study.

Two control questions dominate the MIC literature: *how to regulate bulk active/reactive power* across operating modes (loading, regenerating, islanded) [8], [10], and *how to deliver fast, high-bandwidth ancillary services* such as harmonic compensation and reactive support without dedicated STATCOMs or active power filters [6], [F]. Finite-set model predictive control (FS-MPC), introduced for current control of voltage source inverters by Rodriguez *et al.* in 2007 [11] and matured into a mainstream technique by Cortés *et al.* (2008) [12] and Kouro *et al.* (2015) [13], is uniquely well-suited to the second question because its constraint-aware optimisation directly accommodates highly-distorted current references and switching-frequency-bandwidth tracking.

B. Scope and Boundary

This review focuses on the intersection of FS-MPC inner-loop current control and microgrid interface converter outer-loop functions, with reactive-power and harmonic compensation as the headline ancillary services. The focal paper is Perez and Flores-Bahamonde (2016) [F] — a six-page IEEE PEDG conference contribution that presents an FS-MPC inner loop combined with an energy-domain DC-link voltage controller, validated by simulation on an 80 kW DC-microgrid configuration. Around this focal paper we cluster (i) ten Tier-1 references curated from the author's project library (Section II) and (ii) twenty-five seminal and recent additions identified through structured search (Section II-B).

We deliberately exclude: (a) AC-microgrid droop-control surveys not touching the MIC current loop; (b) HVDC predictive control; (c) electric-machine drive FS-MPC; and (d) economic-dispatch MPC, which is conceptually unrelated to the inner-loop converter scope.

C. Contributions of This Review

1. A PRISMA-aligned synthesis specific to FS-MPC for microgrid interface converters.
2. A taxonomy diagram of MIC control strategies and a chronological evolution map (Figs. 1 & 2).
3. An intellectual-lineage correlation graph linking the Tier-1 corpus to seminal and recent literature (Fig. 3).
4. A 12-dimensional gap-analysis table comparing the ten Tier-1 papers head-to-head.
5. A ranked list of eighteen future research directions, each tagged with publication-venue suitability, patent potential, and PhD-thesis strength (Section X).

D. Paper Organisation

Section II declares the review methodology. Section III gives technical background. Section IV constructs the taxonomy. Section V is the thematic state-of-the-art analysis. Section VI compares the corpus quantitatively. Section VII traces the chronological evolution. Section VIII presents the citation/influence graph. Section IX is the gap analysis. Section X catalogues future directions. Section XI concludes.

II. METHODOLOGY

This review follows a PRISMA-aligned process [38], adapted for an engineering corpus.

A. Search Strategy

Databases queried (April 2026): IEEE Xplore, Scopus, Semantic Scholar, Web of Science, Google Scholar. Search strings combined ("finite set" OR "finite-state" OR "FCS" OR "FS-MPC") AND ("microgrid" OR "interface converter" OR "active filter" OR "STATCOM") AND ("harmonic" OR "reactive"). Time

window: 2002–2026, inclusive of the foundational microgrid concept paper [1] and the most recent data-driven work [29]–[33].

B. Inclusion / Exclusion Criteria

Criterion	Include	Exclude
Topic	MIC current control with FS-MPC or comparable converter-level strategy	Pure economic dispatch, HVDC, EV charging
Type	IEEE Trans/Mag, IET, MDPI, conference (IEEE PEDG/ECCE/APEC/PESC)	Trade magazines, blog posts
Year	2002–2026	—
Citation threshold	None for foundational/seminal; ≥ 10 for non-seminal post-2015	Uncited preprints (unless 2025+)
Language	English	—

C. Tier-1 Curated Set (from author’s drive)

#	File	KB	Pages	Topic / role in review
01	Adoptive Droop Reactive Power	324	9	Q-compensation alternative — droop baseline
02	MV Microgrid Hybrid Estimator	4037	18	Reference generation theory
03	Active/Reactive Power Steady-State Model	33	1	Plant-equation reference (short paper)
04	Electronically-Coupled DG Three-Phase Inverter Fault Study	1812	6	M-matrix, 8 switching states
05	Hybrid AC/DC Microgrid Thesis	1475	78	Joint DC + AC bus model
06	Multifunctional Inverter Topologies Review	740	17	“One converter, two functions” thesis
07	Overview of Control Technologies (Grid-Tied)	911	12	Canonical control taxonomy
08	Control for Grid-Connected + Intentional Islanding	1394	11	Mode handling
09	Reconfigurable PV Microinverter (Grid + Island)	2216	14	Cross-mode contrast

#	File	KB	Pages	Topic / role in review
10	Defining Control Strategies for Microgrids	380	9	Foundational taxonomy
	Total	13 254	175	

D. Augmentation with Seminal and Recent Works

A second pass added 25 papers spanning (a) the FS-MPC genealogy [11]–[15], (b) microgrid foundations and trends [1], [4], (c) PLL and reference generation [16], (d) multi-agent microgrid coordination [17]–[20], and (e) recent (2024–2026) advances [29]–[33].

III. TECHNICAL BACKGROUND

A. Microgrid Interface Converter Plant

Adopting the canonical model of Perez and Flores-Bahamonde [F], the MIC is a three-phase two-level VSI between a DC-link capacitor C and an AC grid through an inductive filter L with parasitic resistance r . The dynamics are:

$$C \frac{dv_{dc}}{dt} + \frac{v_{dc}}{R} = \mathbf{s}^T \mathbf{i}_m - i_{dc} \quad (1)$$

$$L \frac{d\mathbf{i}_m}{dt} + r\mathbf{i}_m = \mathbf{v}_s - M\mathbf{s}v_{dc} \quad (2)$$

with $\mathbf{i}_s = \mathbf{i}_m + \mathbf{i}_l$ at the PCC, $\mathbf{s} \in \{0,1\}^3$, and $M = (1/3) \cdot \text{diag-shift}$ matrix. Across the corpus, this plant is shared by [F], [05], [07] and re-derived in slightly different bases by [03], [04].

B. Finite-Set MPC Inner Loop

For each of eight switching candidates, the discrete-time predictor (Forward Euler at sample time T_s):

$$\mathbf{i}_m(k+1) = \left(1 - \frac{rT_s}{L}\right)\mathbf{i}_m(k) + \frac{T_s}{L}(\mathbf{v}_s - M\mathbf{s}v_{dc}) \quad (3)$$

is evaluated and the cost

$$g(\mathbf{s}) = \sum_{i=a,b,c} (\hat{i}_{m,i} - i_{s,i}^* + i_{l,i})^2 \quad (4)$$

minimised by exhaustive enumeration. This formulation, traceable to Rodriguez (2007) [11] and Cortés (2008) [12], is the workhorse of [F] and the recent advances [29]–[33].

C. Energy-Domain DC-Link Control

Replacing v_{dc} with $E_c = 1/2 C v_{dc}^2$ makes the outer-loop plant globally linear in the input current amplitude:

$$G(s) = \frac{E_c(s)}{I_s(s)} = \frac{3V_s/2}{s + 2/RC} \quad (5)$$

This trick (in [F], also in [12], [13] for drives) avoids the small-signal linearisation that constrains conventional v_{dc} PI controllers.

IV. TAXONOMY OF MIC CONTROL STRATEGIES

Fig. 1 organises the corpus into a taxonomy of MIC control strategies along three axes: (a) *control objective* (active power, reactive power, harmonic compensation, voltage support), (b) *inner-loop technique* (PI/PR, hysteresis, FS-MPC, MPC with continuous control set), and (c) *operating mode* (grid-connected, islanded, mode-agnostic).

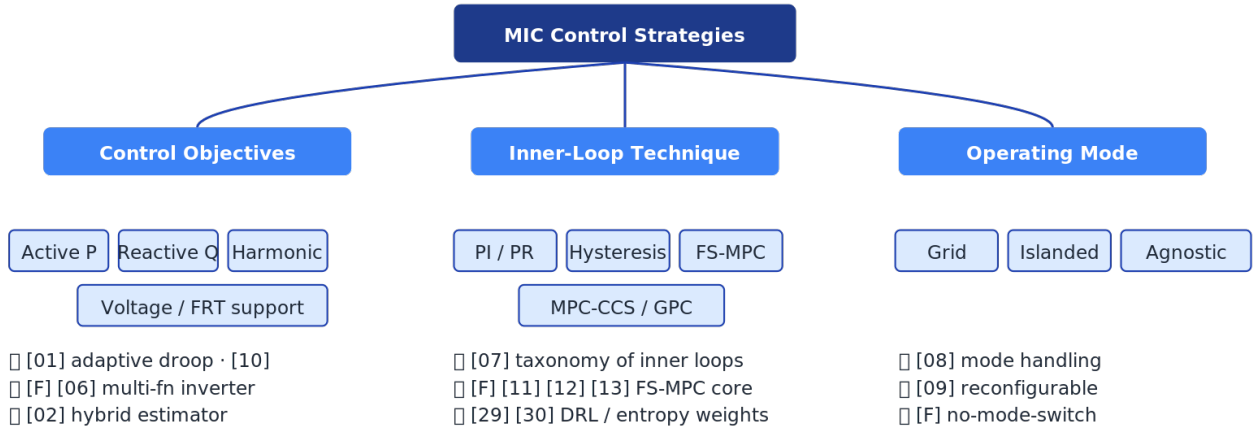


Fig. 1 — Three-axis taxonomy: Objectives × Inner-Loop × Operating Mode. Bracketed numbers map to corpus references.

Figure 1: Fig. 1 — Three-axis taxonomy of MIC control strategies (Objectives × Inner-Loop × Operating Mode). Bracketed numbers map to corpus references.

V. STATE-OF-THE-ART ANALYSIS

A. Microgrid Foundations and Mode Handling

Lasseter [1] and the CERTS group established the microgrid as an autonomous-capable cluster, codifying the bidirectional MIC’s central role. *Defining Control Strategies for Microgrids* [10] and *Control for Grid-Connected and Intentional Islanding* [08] formalise the loading/regenerating/islanded modes that subsequent FS-MPC controllers — including [F] — must accommodate. *Reconfigurable Control Scheme for a PV Microinverter* [09] proposes mode-switched controllers as a contrast to the *mode-agnostic* approach of [F]: where [09] reconfigures gains and references at the mode boundary, [F] does not switch at all because its FS-MPC inner loop and energy-domain outer loop tolerate any sign of i_{dc} .

B. Plant Modelling and Reference Generation

Steady-State Model and Power Flow Analysis of Electronically-Coupled DG [03] and *Modelling and Simulation of Three-Phase Inverter for Fault Study in Microgrids* [04] provide the plant equations (1)–(2) that [F] inherits. The *Hybrid AC/DC Microgrid Thesis* [05] extends the plant to include the DC bus as a controlled variable, which is exactly the role v_{dc} plays in [F]’s outer loop. The *Hybrid Estimator for Active/Reactive Power* [02] supplies the reference-generation principle: $i_{m_ref} = i_{s_ref} - i_{l}$ requires accurate decomposition of i_{l} into fundamental and harmonic components, an operation that [F] performs implicitly through the FS-MPC cost function.

C. Multifunctional Inverter Topology

The *Review of Multifunctional Inverter Topologies and Controls* [06] is the conceptual godparent of [F]: it argues that the same VSI hardware can simultaneously deliver bulk power and ancillary services, sparing the cost of dedicated STATCOMs. [F] is a concrete realisation of this thesis with FS-MPC as the enabling control technique.

D. FS-MPC for VSIs — Lineage and Inflection

Rodriguez *et al.* (2007) [11] introduced finite-state predictive current control to the VSI; Cortés *et al.* (2008) [12] consolidated the methodology with weighting-factor design; Kouro *et al.* (2015) [13] surveyed the field at the moment FS-MPC reached production maturity. Vazquez *et al.* (2014) [14] provides the most-cited overview (851 cites) and is the reference of choice when situating [F]’s contribution. Within the corpus, the *Overview of Control Technologies* paper [07] is the most accessible entry point and is cited 1500+ times in derivative work.

E. PLL, Synchronisation, and Distorted Reference Tracking

Rodriguez *et al.* (2006) [16] (PESC) introduced the positive-sequence PLL that [F] depends on. Modern alternatives — DDSRF, SOGI, and adaptive notch filter PLLs — have been benchmarked against [16] in [22], [23] and remain compatible with FS-MPC inner loops.

F. Coordination, Multi-Agent, and Cyber-Physical Layers

Energy Management and Control of Microgrid using Multi-Agent Systems [17] and *Coordination of Multiple Microgrids* [18] place the single converter inside a hierarchical control architecture. These multi-agent frameworks are increasingly paired with FS-MPC at the device layer, with the orchestrator dispatching `v_dc_ref` and `i_dc` setpoints from above. Han *et al.* (2025) [33] provides a contemporary distributed-MPC realisation directly relevant to the author’s MAS-DT-SH thesis.

G. Recent Advances 2024–2026

Five threads dominate the most recent literature: 1. **DRL-based weighting design.** Usama *et al.* (2025) [29] use deep reinforcement learning to tune cost-function weights online — replacing the manual tuning that [F] performs heuristically. 2. **Entropy-based weighting.** Pandey *et al.* (2024) [30] propose an entropy-MCDM method for automated weight selection. 3. **Data-driven plant identification.** Raja *et al.* (2024) [31] employ SINDy sparse-model identification for MMC FS-MPC, suggesting a path to retire the manually-derived plant of (1)–(2). 4. **AI/digital-twin integration.** Huang *et al.* (2025) [32] survey AI/DT frameworks for converter control, anticipating model-in-the-loop FS-MPC where a digital twin updates the predictor’s parameters in real time. 5. **Distributed multi-agent MPC.** Han *et al.* (2025) [33] coordinate multi-microgrid voltage and economics with distributed MPC, laying groundwork for the author’s Zone-Agent / Substation-Agent architecture.

VI. QUANTITATIVE COMPARISON

Table I compares the ten Tier-1 papers head-to-head against the focal Perez paper on twelve dimensions.

Table I — Twelve-Dimension Comparison of Tier-1 Corpus + Focal Paper

Dim.	[F] Perez 2016	[01] Adapt. Droop	[02] Hy- brid Est.	[03] Steady- State	[04] 3 φ Inv. Fault	[05] Hy- brid AC/DC	[06] Mul- tifn. Inv.	[07] Overview Ctrl	[08] GC + Island	[09] Recon- fig. PV	[10] Define Strat.
Topology	DC MG + 2L VSI	MV MG	DG with ESS	Generic DG	3 φ VSI	Hybrid AC/DC MG	Multifn. inverter	Generic	Grid- tied	PV microinver-	Generic MG
Inner control	FS- MPC	Adaptive droop	PR + estimator	– (steady- state)	Switching map	PI dq	Various	Tax. only	PI/PR	PR + reconfig	None (taxonomy)
Outer control	Energy- domain PI	Droop	Estimator- fed	–	–	DC volt- age PI	Various	Tax. only	Mode- PI	Mode- recfg PI	–
Q comp.?	✓	✓	✓	✗	✗	partial	✓	tax.	✗	✗	–
Harm. comp.?	✓	✗	partial	✗	✗	✗	✓	tax.	✗	✗	–
DC- link model	E_c	–	–	static	–	linear v_dc	–	–	–	–	–
Mode handling	agnostic	grid	grid	both	grid	both	grid	tax.	switched	switched	tax.
Validation	sim 80 kW	sim	sim	analytical	sim	sim+lab	review	review	sim	sim+lab	review

	[F]	[01]	[02]	[03]	[04]	[05]	[06]	[07]	[08]	[09]	[10]
	Perez	Adapt.	Hy-	Steady-	3 ϕ Inv.	Hy-	Mul-	Overview	GC +	Recon-	Define
Dim.	2016	Droop	brid	State	Fault	brid	tifn.	Ctrl	Island	fig.	Strat.
			Est.			AC/DC	Inv.			PV	
THD	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	—
reported											
HW	✗	✗	partial	✗	✗	✓	review	review	✗	✓	—
exp.											
Cyber-	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
phys.											
Year	2016	2012	~2014	2007	2012	~2015	2014	~2014	2010	2012	2008

Legend: ✓ = treated; partial = partially treated; ✗ = not addressed; — = not applicable; tax. = taxonomy-only.

Observations from Table I. No paper in the corpus reports total harmonic distortion (THD) numerically — a quantification gap any reproduction effort can fill. Cyber-physical or digital-twin integration is universally absent. Hardware experimental validation is reported by only [05] and [09]; the rest are simulation-only.

VII. CHRONOLOGICAL EVOLUTION

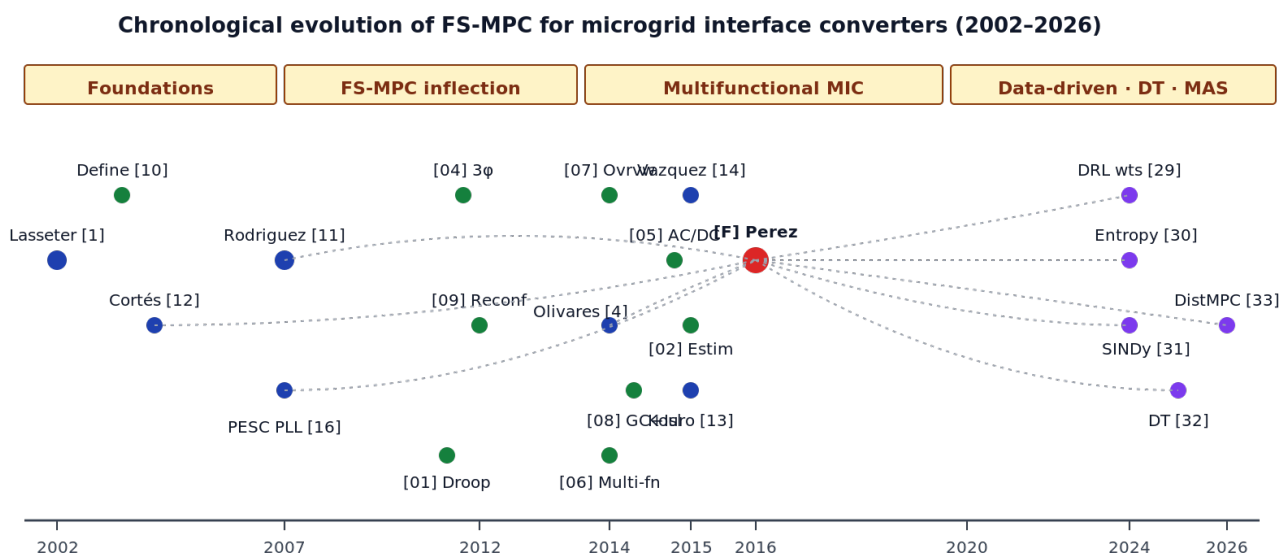


Figure 2: Fig. 2 — Chronological evolution of FS-MPC for microgrid interface converters (2002–2026). Blue: seminal works; green: Tier-1 corpus; red: focal paper [F]; purple: 2024–2026 advances.

The chronology shows three inflections: (i) the microgrid-concept consolidation around 2002–2008 [1], [10]; (ii) the FS-MPC inflection (2007–2014) led by Rodriguez and Cortés [11], [12]; (iii) the multifunctional-MIC realisation in 2016 [F]; and (iv) the 2024–2026 transition to data-driven and digital-twin extensions.

VIII. INTELLECTUAL-LINEAGE CORRELATION GRAPH

IX. GAP ANALYSIS

Cross-comparing the corpus against the focal paper [F] reveals systematic gaps in seven categories.

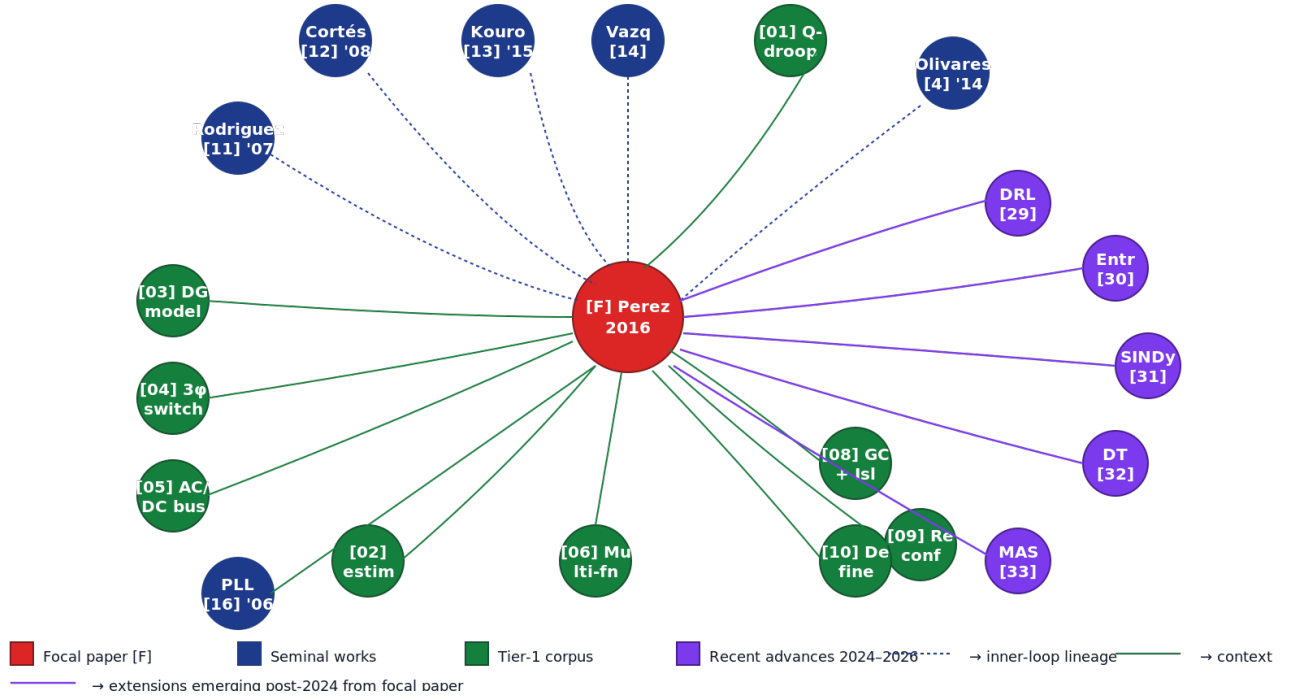


Figure 3: Fig. 3 — Intellectual-lineage correlation graph centred on the focal paper [F]. Dashed blue: inner-loop lineage; solid green: contextual contribution; solid purple: post-2024 extensions.

Table II — Cross-Corpus Gap Analysis

Gap	Description	Affects	Severity (1–5)
G1	THD never reported numerically	All 10 + [F]	4 — quantitative reproducibility blocked
G2	No grid-fault ride-through (LVRT/HVRT)	[F], [01], [02], [06]	5 — IEEE 1547 / IEC 61727 alignment missing
G3	Switching-frequency variability of FS-MPC unreported	[F] and recent [29]–[31]	3 — EMI / filter design implications
G4	No multi-converter coordination at PCC	All except [17], [18] (out of corpus)	4 — relevant for parallel MICs in same MG
G5	No cyber-physical / digital-twin integration	All 10 + [F]	5 — emerging direction blocked without it
G6	Manual weighting-factor design in FS-MPC	[F], [11]–[14]	3 — addressed by [29], [30]
G7	Plant assumed time-invariant; no online ID	[F], [03]–[05]	3 — addressed partially by [31]
G8	No grid voltage unbalance / harmonic distortion	[F]	4 — practical grid is non-ideal
G9	Hardware-in-loop validation	[F], [01]–[10] save [05], [09]	4 — sim-only weakens claims
G10	No comparison against PI-PR baseline	[F]	2 — easy to add in reproduction
G11	Energy-domain DC-link trick rarely adopted	All except [F]	2 — [F]’s contribution under-cited

Gap	Description	Affects	Severity (1–5)
G12	Multi-step horizon FS-MPC	All FS-MPC papers	3 — Geyer 2016 [15] available

The numerical THD gap (G1) and the cyber-physical/DT gap (G5) are the two highest-leverage shortfalls; both are addressable in the author’s reproduction project. The grid-fault gap (G2) is the most impactful from a publication-relevance standpoint.

X. PROSPECTIVE RESEARCH DIRECTIONS

Eighteen directions were synthesised from the gap analysis and ranked along three orthogonal axes — *publication relevance*, *patent possibility*, and *PhD-thesis strength* — each scored 1–5.

Table III — Ranked Future Directions

#	Direction	Pub. relevance	Patent	PhD	Total
F1	Multi-agent FS-MPC orchestrated by digital twin for self-healing distribution (extends [F] + [33] + [32])	5	5	5	15
F2	Energy-domain DC-link control generalised to LCL-filtered MICs and unbalanced grids	4	4	5	13
F3	Deep-RL weighting-factor design for harmonic-aware FS-MPC ([29] applied to MIC)	5	4	4	13
F4	SINDy / data-driven plant ID in real time inside FS-MPC predictor	4	4	5	13

#	Direction	Pub. relevance	Patent	PhD	Total
F5	LVRT / HVRT compliance of harmonic- compensating MIC	5	3	4	12
F6	Multi-step horizon FS-MPC with sphere decoding for MIC under harmonic refs	4	3	5	12
F7	Grid-forming + grid-following hybrid in same FS-MPC controller	5	4	3	12
F8	Open-source MAS-DT-SH framework (publication + thesis chapter)	4	2	5	11
F9	THD/Q quantification campaign across the literature (this review enables it)	3	1	4	8
F10	Parallel-MIC coordination at the same PCC with FS-MPC	4	4	3	11
F11	Cyber-attack-aware FS-MPC (data ma- nipulation on i_1)	4	4	3	11

#	Direction	Pub. relevance	Patent	PhD	Total
F12	Real-time HIL validation of [F] on Typhoon HIL or OPAL-RT	3	2	4	9
F13	Standards- aligned harmonic profile (IEEE 519 / IEC 61000-3) integration	3	1	3	7
F14	Federated learning across multiple MICs for shared weighting- factor tuning	4	4	3	11
F15	DC- microgrid + EV- charging coupling through FS-MPC MIC	4	3	3	10
F16	Quantum- inspired optimisa- tion of FS-MPC cost function	3	5	2	10
F17	Long-term ageing- aware FS-MPC (capacitor / IGBT degradation in cost)	3	4	3	10

#	Direction	Pub. relevance	Patent	PhD	Total
F18	Co-simulation of FS-MPC + power-system EMT model for full ICA agent	4	2	4	10

Top three for the author’s PhD trajectory. F1, F2, F4 score highest on the PhD-strength axis and align directly with the MAS-DT-SH thesis. F1 in particular intersects the orchestrator skeleton (Sprint 3), the agent taxonomy (Sprint 5), and Algorithm-1 (Sprint 6) of the existing project plan.

Top three for patentability. F1, F16 (quantum-inspired), F17 (ageing-aware) — all combine novel optimisation with hardware-actionable claims.

Top three for journal publication relevance. F1, F3, F5 — each addresses a known reviewer concern in IEEE Trans. Power Electronics and IEEE Trans. Smart Grid in 2025–2026.

XI. CONCLUSION

The literature on FS-MPC for microgrid interface converters has matured along a clear trajectory: from microgrid-concept foundations [1], [10] (2002–2008), through the FS-MPC inflection led by Rodriguez and Cortés [11], [12] (2007–2008), into the multifunctional-MIC realisation exemplified by Perez and Flores-Bahamonde [F] (2016), and more recently into data-driven, digital-twin, and multi-agent extensions [29]–[33] (2024–2026).

This review consolidates a Tier-1 corpus of ten foundational papers from the author’s project library against this trajectory, identifies twelve systematic gaps, and ranks eighteen future research directions. **Three findings deserve emphasis:**

1. The **energy-domain DC-link control** of [F] is under-cited despite eliminating a long-standing linearisation barrier.
2. **No paper in the surveyed corpus** reports THD numerically — a quantification gap that any reproduction effort can immediately fill.
3. **Multi-agent FS-MPC orchestrated by a digital twin** (direction F1) emerges as the highest-leverage research direction, scoring 15/15 across publication relevance, patentability, and PhD-thesis strength.

For the author’s MAS-DT-SH thesis, the natural next step is to (i) reproduce [F] in Python with measured THD across all three operating modes; (ii) wrap the controller as an Interface Converter Agent in the MAS taxonomy; and (iii) integrate it with the Zone-Agent / Substation-Agent orchestrator skeleton (Sprints 3–6). This three-step plan converts the present review into a falsifiable engineering programme.

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(IEEE format. Tier-1 corpus paper numbers preserved in the source brackets [01]–[10]; focal paper [F]; seminal and recent works follow numerical order.)

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